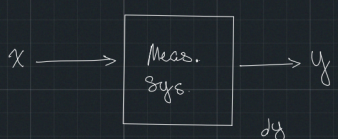
9/30 Bhargav T 2:30 - 3:30 Baker 1120
Yen-Chun W 11:30 - 12:30
Eli R 2:30 - 3:30

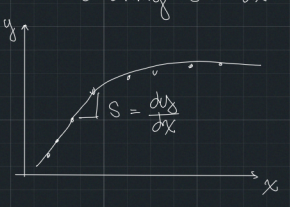
TEMP MEASUREMENT (ch 8 in text)

On the lab apparatus: Bulb thermometers

Other devices: Thermistors
RTDs
Thermocouples



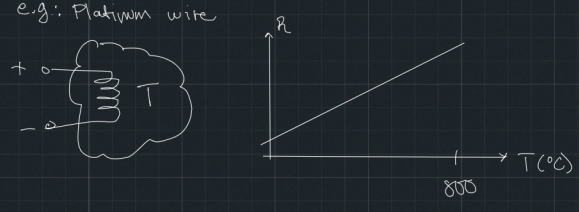
Sensitivity S: $\frac{dy}{dx}$



For large $|S|$, small changes in x can be measured. (Noise may be an issue)
In contrast, small $|S|$ makes it more difficult to to measure small changes in x

RTD (Resistance Temp Detector) — can measure T to a precision of $\pm 0.005\,^\circ C$
— Thin wires may break easily
Exploits the fact that the electrical resistance of a material depends on its temp.

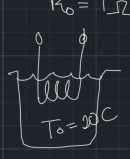
e.g.: Platinum wire



$R_t = R_0 + R_0\alpha(T - T_0)$; $R \equiv$ Resistance at T
$R_0 \equiv$ Resistance at T_0 (Reference temperature)
$\alpha \equiv$ Temp. coefficient of sensitivity (Material property)

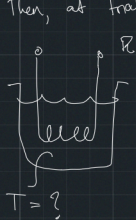
If at trial 1:

$R_0 = 1\,\Omega$



$\alpha_{Pt} = 0.00392\,^\circ C^{-1}$

Then, at trial 2:



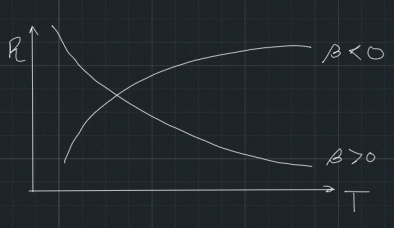
$R = R_0 + R_0\alpha_{Pt}(T - T_0)$ ← solve for T

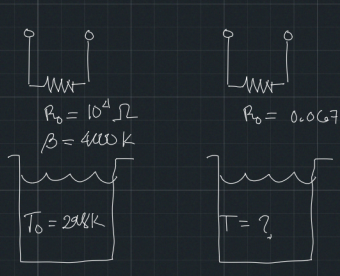
$S \equiv \frac{dy}{dx} = \frac{dR}{dT} = R_0\alpha_{Pt}$

Thermally sensitive resistor (Thermistor)

— Based on semiconductors
— Highly nonlinear
— Rugged
— May fail at high T ($\gtrsim 250^\circ C$)
— precision of $\pm 0.01\,^\circ C$

$R_t = R_0 e^{\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)}$; $\beta \equiv$ Material property where $3700\,K \lesssim |\beta| \lesssim 4600\,K$



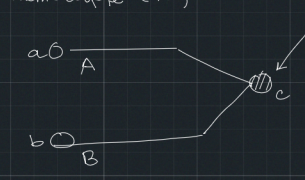


Solve for T: $R = R_0 e^{\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)}$

$S \equiv \frac{dy}{dx} = \frac{dR}{dT} = R_0 e^{-\frac{\beta}{T_0}} e^{\frac{\beta}{T}}$

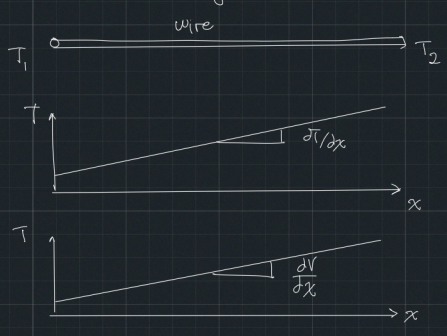
$= \boxed{R_0 e^{-\frac{\beta}{T_0}} e^{\frac{\beta}{T}} \frac{\beta}{T^2}}$

Thermocouple (TC)



If locations a, b, and c are at different temps, then a voltage diff. $\Delta V = V_b - V_a$ will develop

Seebeck effect: A voltage difference will develop between two points of an electrically conducting material if there is also a temp diff between these points



$S_A = \frac{\frac{dV}{dx}}{\frac{dT}{dx}} = \frac{dV}{dT} \equiv$ Seebeck coeff

Generally, $S = S(T)$ where $S = \frac{dV}{dT} \Rightarrow dV = S\,dT$

$\int_{V_1}^{V_2} dV = \int_{T_1}^{T_2} S\,dT$

$V_2 - V_1 = \Delta V = \int_{T_1}^{T_2} S\,dT$

Often S is treated as a constant
Then $V_2 - V_1 = S(T_2 - T_1)$

Typical Seebeck coeff:
Cu at 20°C: $S_{Cu} = 9\,\frac{\mu V}{^\circ C}$